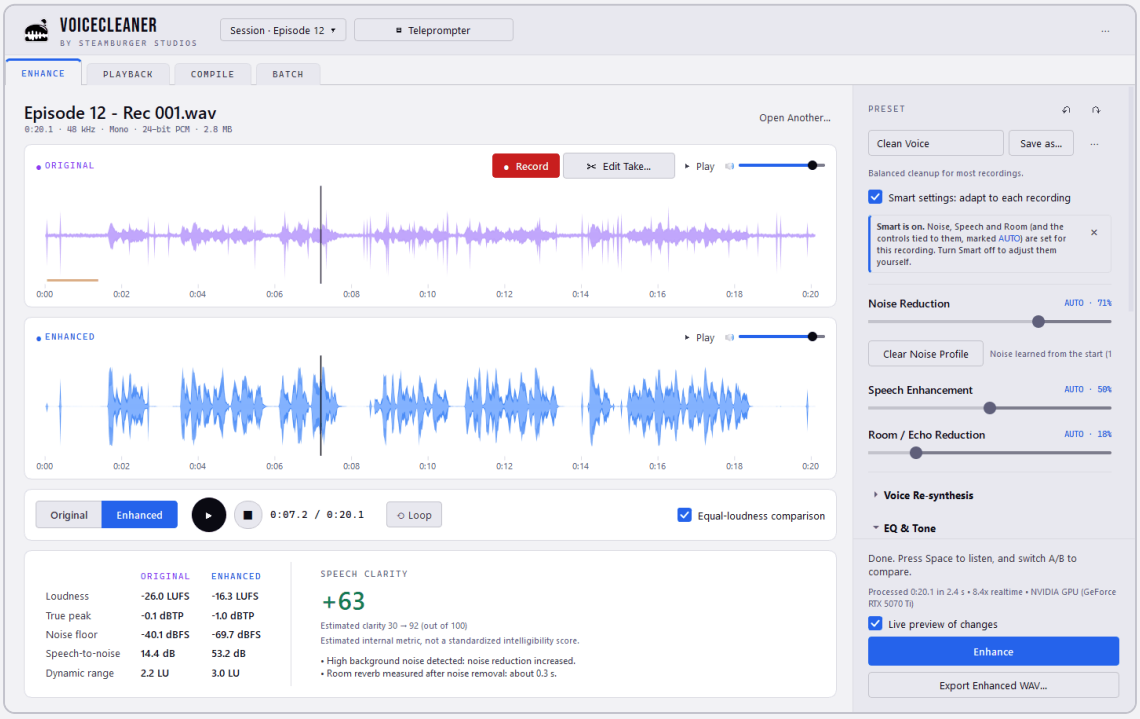
STEAMBURGER STUDIOS

VOICECLEANER

USER MANUAL

Clean, clear speech from noisy recordings: free, offline, for Windows and Mac. Quick start, every tab and setting, keyboard shortcuts and troubleshooting.



CONTENTS

01 Welcome 3

02 Quick Start 4

1. Install VoiceCleaner 4
2. Start it and take the tour 4
3. Open a recording 5
4. Pick a preset 6
5. Enhance and listen 7
6. Export 8
Bonus: clean a whole folder 9

03 The Window at a Glance 10

04 Sessions 12

05 The Enhance Tab 13

06 Settings and Presets 16

07 Recording 19

08 The Teleprompter 21

09 The Compile Tab 23

10 The Batch Tab 25

11 The Playback Tab 26

12 Export Formats 27

13 Settings 28

14 Command Line 30

15 Keyboard Shortcuts 31

16 Updates 32

17 Reference 33

Supported input 33
How VoiceCleaner cleans speech 33
Files and folders 34

18 Troubleshooting 35

19 License and Credits 36

CHAPTER 01

WELCOME

VoiceCleaner turns noisy, echoey or uneven speech recordings into clean, clear audio. It runs entirely on your own computer: no account, no subscription, no upload.

Open a recording (a phone memo, a Zoom call, a podcast take, the sound of a video) and VoiceCleaner removes background noise, room echo and hum, evens out the level and makes the voice clear. You compare before and after with an instant A/B switch, then export.

It is built for two jobs:

- **Cleaning recordings you already have**, one at a time or a whole folder at once in the Batch tab.
- **Recording new material**: single takes, whole sessions, or a script read from the built-in teleprompter, which you then join into one file in the Compile tab.

> YOUR AUDIO STAYS WITH YOU
>
> Everything is processed on your computer. Nothing about you, your computer or your audio is sent anywhere. The only network requests VoiceCleaner makes are an optional update check and, if you ask for it, a one-time download of the voice re-synthesis model.

HOW TO USE THIS MANUAL

If you just want a clean file, read the Quick Start: it takes about five minutes. The chapters after it explain each tab and setting in detail, and the last chapters are reference material: keyboard shortcuts, the command line, file locations and troubleshooting.

Names of buttons and menu items are written in `this style`, keys like `Space`, and file names like `Episode 12 - Rec 001.wav`. Where Windows and Mac differ, both are given; `Ctrl` is `⌘ Cmd` on a Mac.

CHAPTER 02

QUICK START

From download to a clean, exported file in six steps.

1 Install VoiceCleaner

Download the version for your computer from the latest release page on GitHub:

YOUR COMPUTER	DOWNLOAD
Windows 10 or 11, any PC	`VoiceCleaner-<version>-win64.msi`
Windows with an NVIDIA graphics card (much faster)	`VoiceCleaner-GPU-<version>-web.exe`[1]
Mac with Apple Silicon (M1 or newer), macOS 12+	`VoiceCleaner-<version>-macos-arm64.dmg`

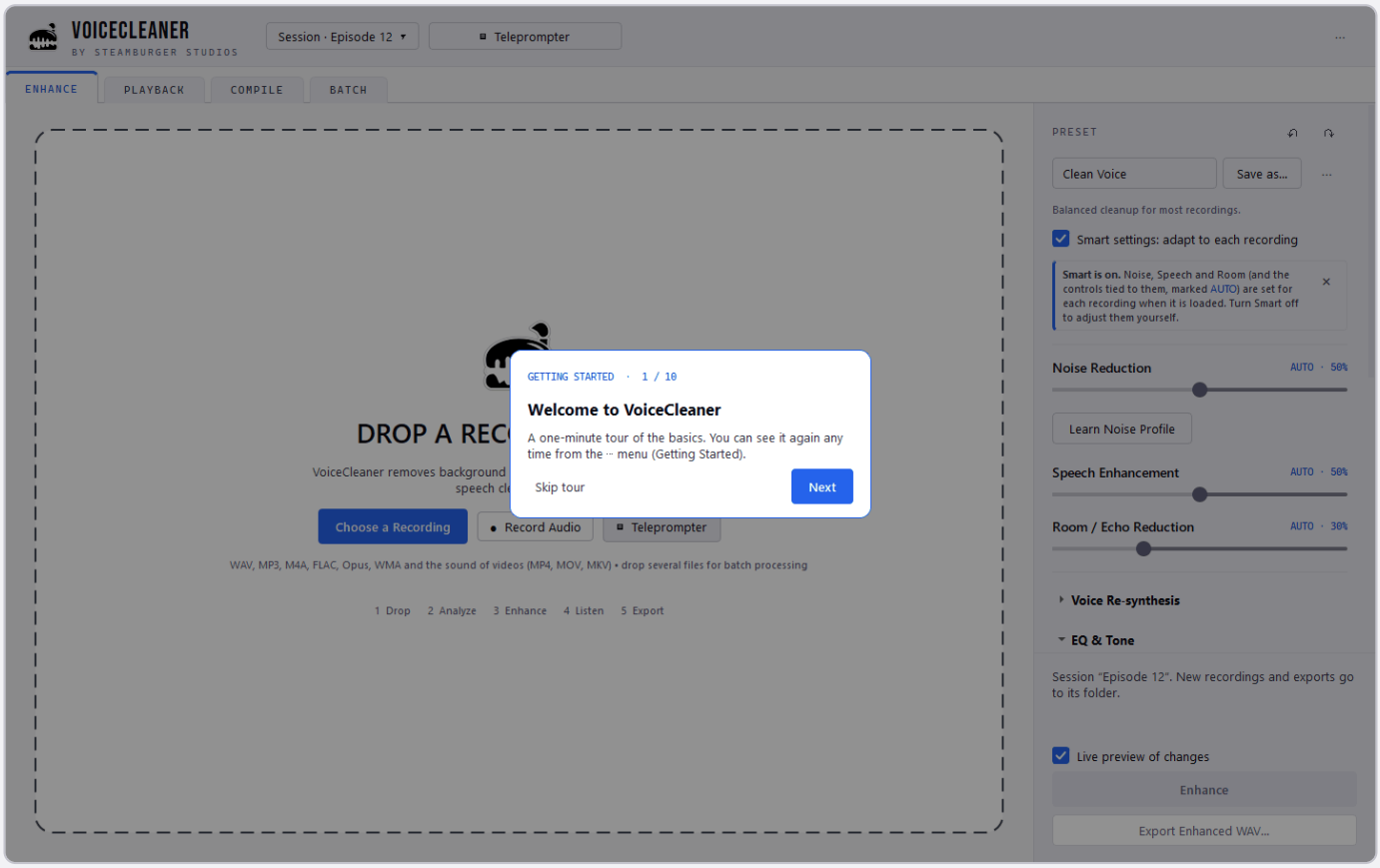
Run the installer (on a Mac, drag VoiceCleaner to Applications). Portable versions that need no installation are on the same page. VoiceCleaner needs 8 GB of memory.

> **"WINDOWS PROTECTED YOUR PC"**
>
> The downloads are not code-signed yet, so Windows may show this warning the first time.[2] Choose `More info` → `Run anyway`. On a Mac, macOS may ask you to confirm opening it the first time.

2 Start it and take the tour

A splash screen appears immediately and shows what is loading; the window opens once the AI engine is ready. The first time, a one-minute **Getting Started** tour points out the basics, one stop per tab. You can see it again any time from the `⋯` menu → `Getting Started`.

1. The GPU edition is a web installer: it downloads about 2 GB of NVIDIA CUDA libraries during installation, so it needs an internet connection the first time.
2. Code signing is a paid certificate that tells Windows SmartScreen who published a program. Without it, SmartScreen warns about any new download until it has seen enough installs. The installer is the same either way.

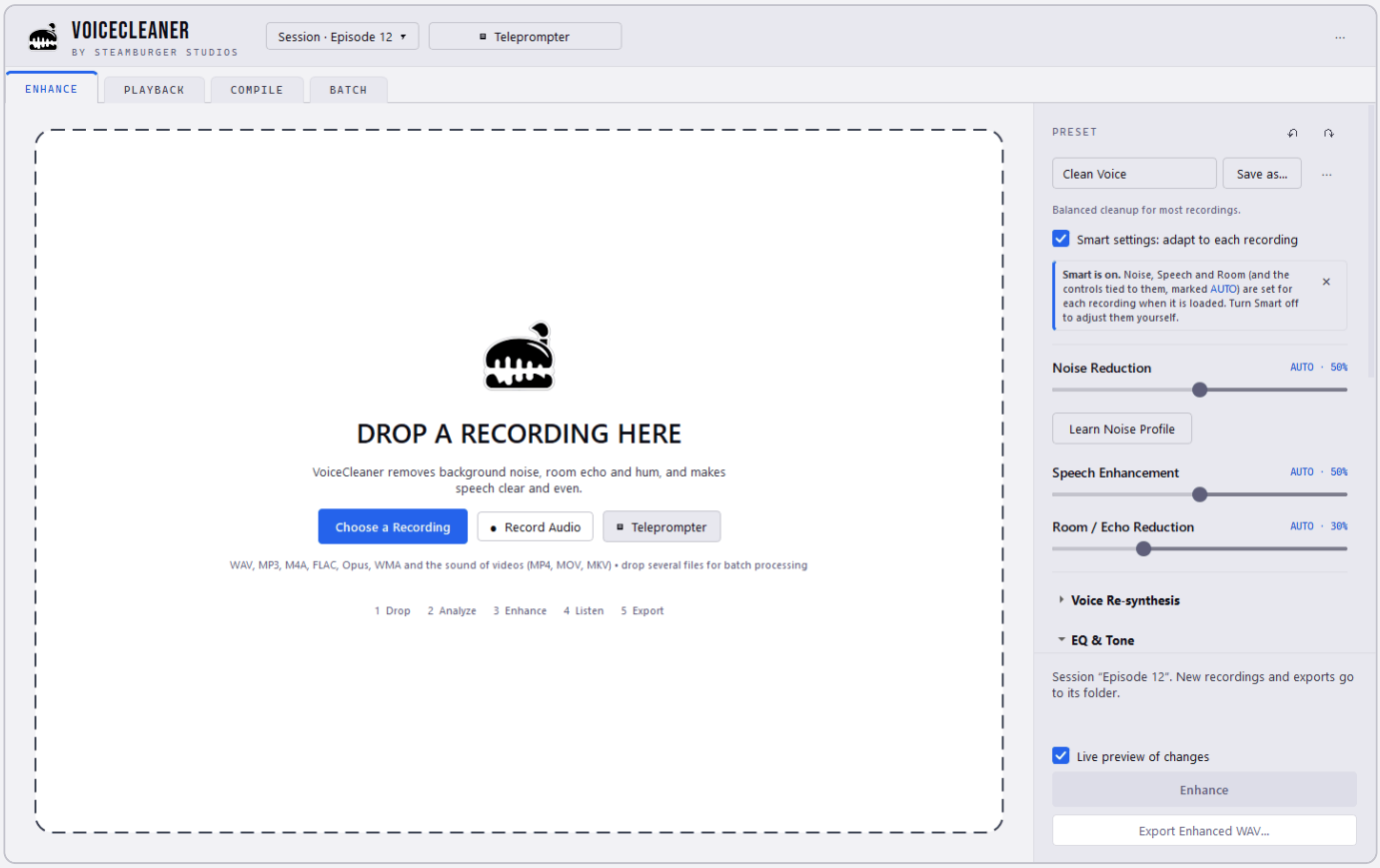


The Getting Started tour dims the window and explains one control at a time. Use Next, Back or Skip.

3 Open a recording

Drop a file on the window, click `Choose a Recording`, or press `Ctrl`+`O`. WAV, MP3, M4A, FLAC, Opus, AIFF, WMA and the sound of video files (MP4, MOV, MKV, WebM) all work.[3] The start screen also lists your `RECENT` files and the newest recordings of the current session for one-click access.

3. M4A, AAC, WMA, AMR and video files are decoded by the LGPL build of FFmpeg included in the Windows and Mac downloads; WAV, MP3, FLAC, Opus/OGG, AIFF and CAF are opened directly. See Supported input.



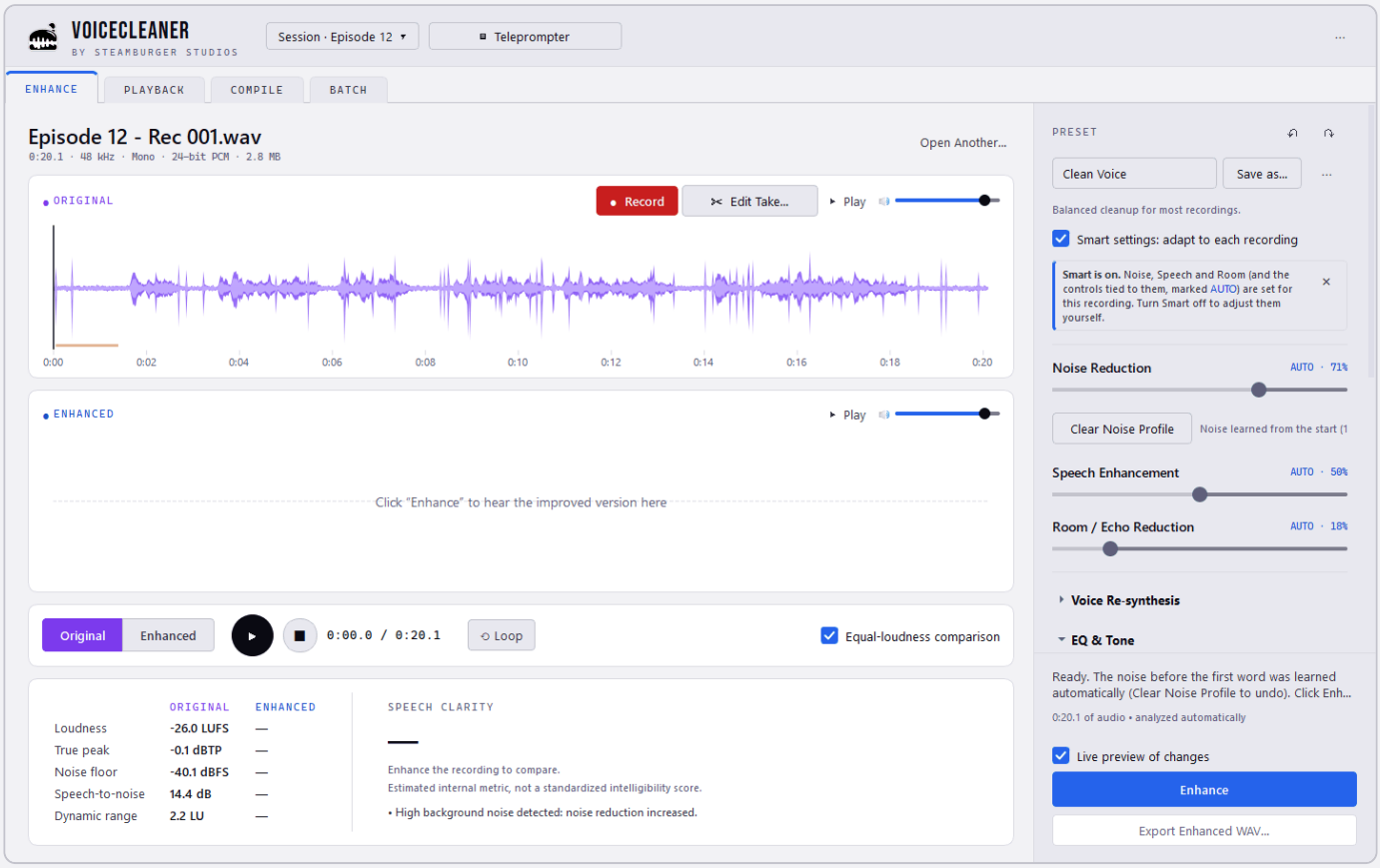
▸ The start screen. Drop a recording, choose one, record, or open the teleprompter.

VoiceCleaner analyzes the recording as it opens: the noise, the room, hum, clipping and the level. If the recording starts with a moment of background noise before the first word, it learns that noise automatically.[4]

4 Pick a preset

The settings sidebar on the right starts on the **Clean Voice** preset with **Smart settings** on. Smart settings uses the analysis to set Noise Reduction, Speech Enhancement and Room / Echo Reduction for this recording; those controls are marked `AUTO`. For most recordings you can go straight to Enhance.

4. A third of a second of steady noise before the first word is enough. The learned profile is shown next to the button, under Noise Reduction; `Clear Noise Profile` undoes it.

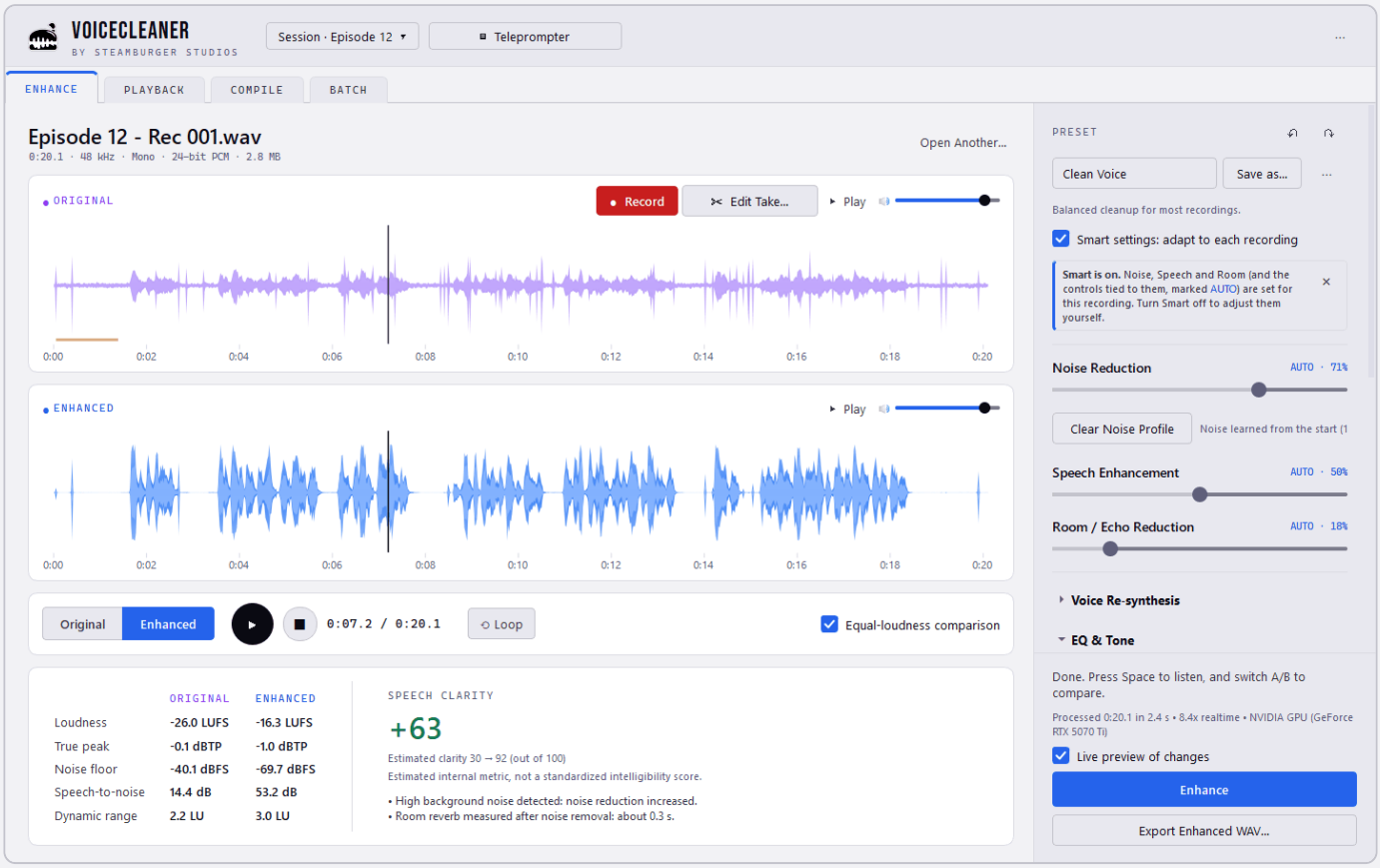


A recording opened and analyzed, ready to enhance. The Original track is purple; the Enhanced track fills in after Enhance.

If the recording is very noisy, choose **Noisy Recording** or **Extreme Noise** instead; for a podcast voice, **Podcast**. All presets are described in Presets.

5 Enhance and listen

Click `Enhance` (or press `Ctrl`+`Enter`). Progress shows on the waveform and the sidebar shows the time left. When it is done, press `Space` to play, and `A` / `B` to switch between the original and the enhanced version at the same position.



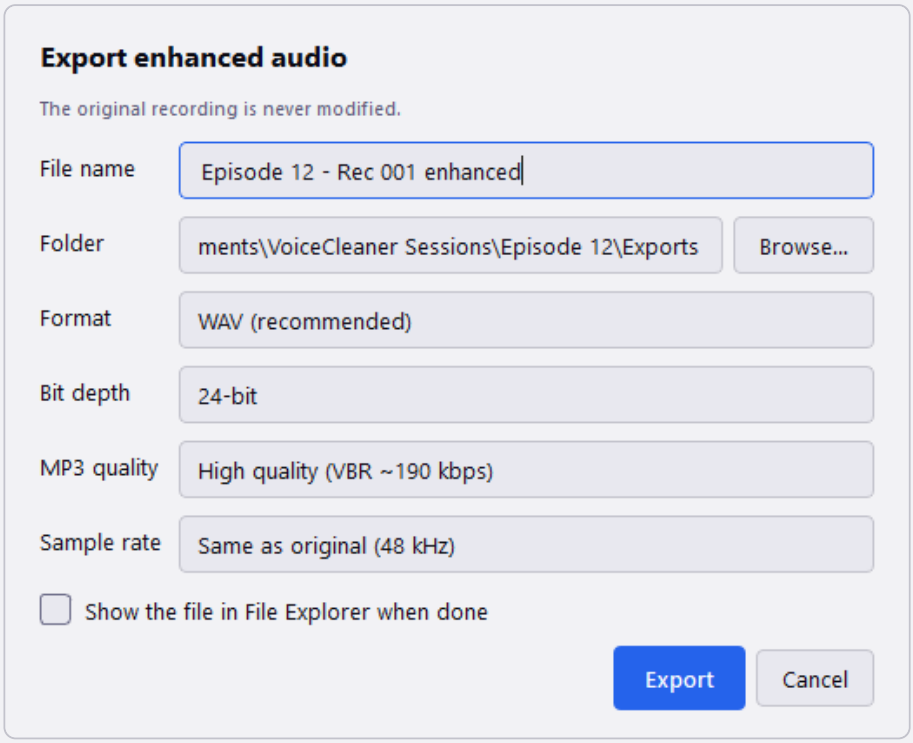
After Enhance: both waveforms, the A/B switch, and before/after measurements with a Speech Clarity score.

Leave `Equal-loudness comparison` on: it plays the original at the enhanced loudness, so you judge the quality, not the volume.[5] If you change a setting now, VoiceCleaner previews the change around the playhead; `Enhance Again` applies it to the whole recording.

6 Export

Click `Export Enhanced WAV…` (or press `Ctrl` + `E`). The file is named after the original and saved to the session's `Exports` folder unless you choose another one. The original recording is never changed.

5. Louder almost always sounds “better” to the ear. Matching loudness removes that bias from the comparison.



The export window: file name, folder, format, bit depth and sample rate.

That is the whole workflow. To clean many files the same way, see the next page.

Bonus: clean a whole folder

1. Open the `BATCH` tab and drop several files, or a whole folder, onto the window.
2. Optional: turn on `◉ Preview` (`P`) and click a file to hear a 10-second sample with the current settings. Adjust the sidebar until it sounds right.
3. Choose where to save under `Save to`, then click `Enhance All`.
4. When it finishes, `Open Output Folder` shows the results.

QUICK-START KEYS

`Ctrl`+`O`	Open a recording	`Space`	Play / pause
`Ctrl`+`Enter`	Enhance	`A` / `B`	Original / Enhanced
`Ctrl`+`E`	Export	`R`	Record / stop
`Ctrl`+`T`	Teleprompter	`Ctrl`+`/`	All shortcuts

CHAPTER 03

THE WINDOW AT A GLANCE

One header, four tabs, and a settings sidebar shared by Enhance and Batch.

THE HEADER

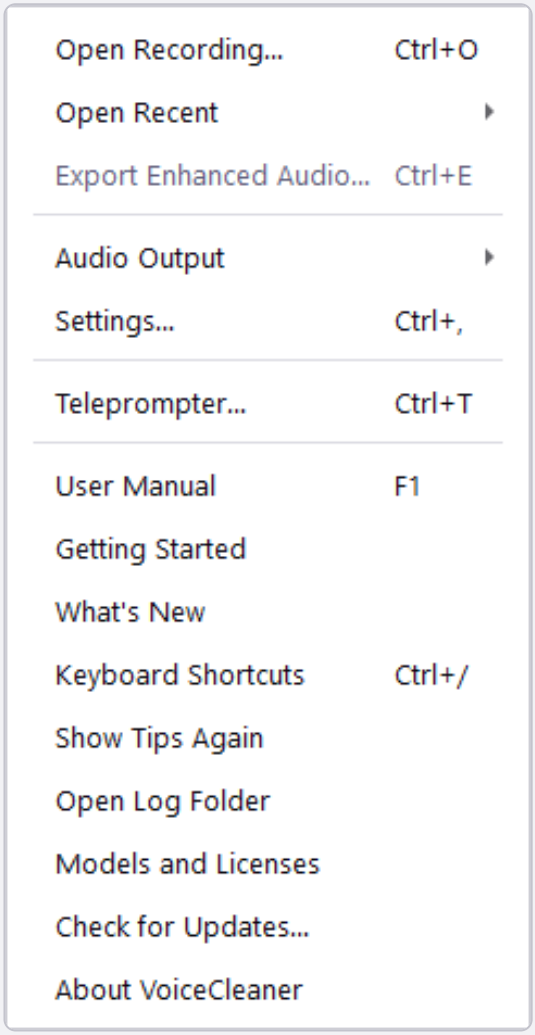
- **Session**: the current session (project). Click it for the session menu; see Sessions.
- **Teleprompter**: opens the teleprompter. It shows when the teleprompter is open, and `● Recording` while it records.
- **Activity chip**: while something runs (enhancing, a batch, a preview), a chip shows it from any tab. Click it to go there.
- **Device and AI model**: shown while the AI engine loads, and when something needs your attention: no AI model, the NVIDIA GPU chosen but not in use, or a faster edition available for this PC (for example `CPU · RTX 5070 Ti found`). Otherwise it is in Settings.
- ⋯: the app menu.

THE TABS

TAB	WHAT IT IS FOR
`ENHANCE`	Open or record one recording, enhance it, compare A/B, export. Chapter 5.
`PLAYBACK`	Browse and play the session's recordings and exports, or any folder. Chapter 11.
`COMPILE`	Join enhanced takes into one file, with trimming and editing. Chapter 9.
`BATCH`	Enhance many files with the same settings. Chapter 10.

A tab that is busy carries a badge (for example `BATCH · 3/10`). On Windows the taskbar button shows progress, and flashes when a long job finishes while VoiceCleaner is in the background. VoiceCleaner reopens on the tab you used last.

THE ⋯ MENU



The app menu.

- `Open Recording…`, `Open Recent` (the last ten files), `Export Enhanced Audio…`
- `Audio Output`: where all playback goes. See Settings.
- `Settings…` (`Ctrl`+`,`): appearance, contrast, fonts, processing device and update checks.
- `Teleprompter…`
- `User Manual` (`F1`), `Getting Started`, `What's New`, `Keyboard Shortcuts`, `Show Tips Again`
- `Open Log Folder`, `Models and Licenses`, `Check for Updates…`, `About VoiceCleaner`

On a Mac, every command is also in the menu bar (File, Edit, Playback, View, Window, Help), with its keys; `About`, `Settings…` and `Check for Updates…` are in the VoiceCleaner menu.

Quitting while VoiceCleaner is working (enhancing, exporting, running a batch or recording) asks first.

CHAPTER 04

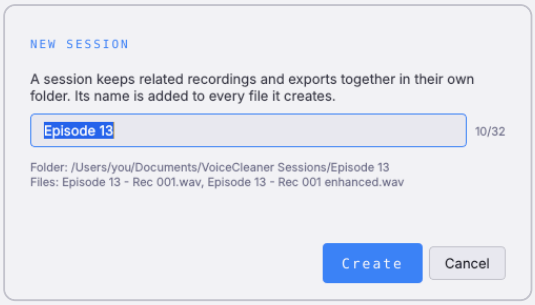
SESSIONS

A session is a project folder: its recordings and exports stay together.

Start a new session for each episode, client or project. VoiceCleaner resumes the last session when it starts.



▸ The session menu, from the SESSION button in the header.



▸ Naming a new session. The preview shows how its files will be named.

- **New Session…** (`Ctrl`+`Shift`+`N`) creates `Documents\VoiceCleaner Sessions\<name>\` with a `Recordings` and an `Exports` folder. A new session starts on the Enhance tab with nothing open; if the open recording was enhanced but not exported, VoiceCleaner offers to export it first.
- **Open Session…** switches to another session.
- **Rename Session…** renames the session and every file it created.
- **Open Session Folder** and **Open Default Recordings Folder** show them in Explorer or Finder.

Every file a session creates carries its name, so files stay identifiable wherever they end up:

```
Episode 12 - Rec 003.wav           a recording
Episode 12 - Rec 003 enhanced.wav  its enhanced export
Episode 12 - Rec 004 append.wav    a take appended to it
Episode 12 - Compilation.wav       the Compile tab's result
```

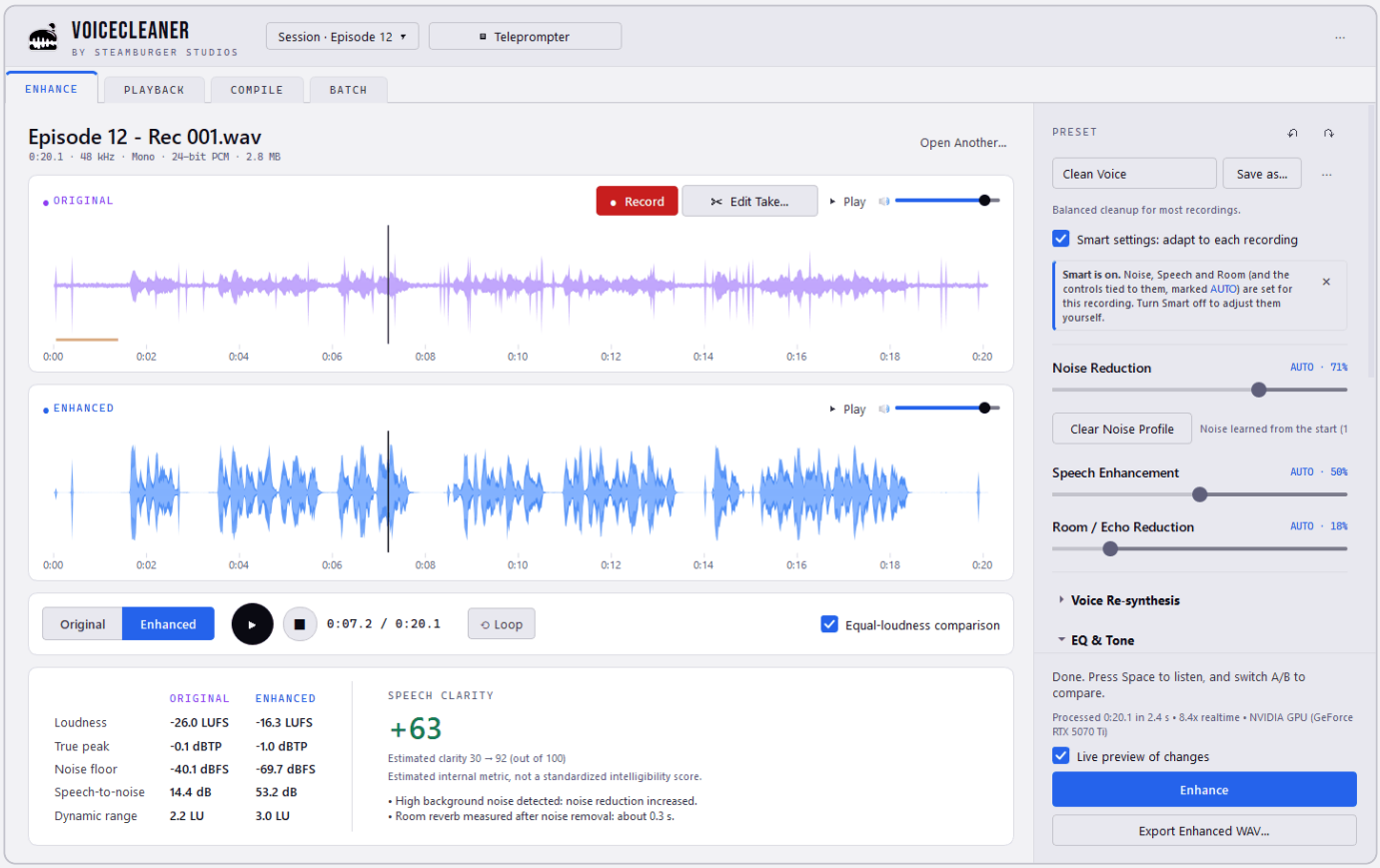
Session names are limited to 32 characters and cleaned of characters Windows does not allow in file names.[6]

6. Those are `\ / : * ? " < > |`. Keeping names short keeps full file paths well under the limits some older Windows tools still have.

CHAPTER 05

THE ENHANCE TAB

Two waveforms, one transport, measurements, and the settings sidebar.



The Enhance tab after enhancing.

WAVEFORMS

The `ORIGINAL` track (purple) shows the recording as it is; the `ENHANCED` track (blue) shows the result. Each has its own `Play` button and volume. Click a waveform to move the playhead; `Ctrl`+mouse wheel zooms, and both tracks stay in step.

Selecting: drag across either waveform to select a passage. The playhead jumps to the start of the selection. A selection is used by `Loop`, by `Learn Noise Profile`, and by Live Preview.

TRANSPORT AND A/B

- `Original` / `Enhanced` (`A` / `B`): the switch is instant and click-free at the same position.
- ▶ / ■ and `Space` play and stop; `Home` goes back to the start.
- `Loop` (`L`) repeats the selection (or the whole recording) while you switch A/B or change settings.
- `Equal-loudness comparison` plays the original at the enhanced loudness.

- `Learn Noise Profile`, in the sidebar under Noise Reduction: select a part with only background noise (no speech) and click it. A learned profile is trusted, so that noise is removed more firmly. `Clear Noise Profile` undoes it.

MEASUREMENTS

Under the transport, VoiceCleaner shows the original and enhanced values side by side:

MEASUREMENT	WHAT IT MEANS
Loudness	Integrated loudness in LUFS.[7] VoiceCleaner targets -16 LUFS by default.
True peak	The highest peak, including peaks between samples, in dBTP.[8]
Noise floor	How loud the background is between words. Lower is quieter.
Speech-to-noise	How far the voice stands above the background, in dB. Higher is cleaner.
Dynamic range	How much the level varies, in LU.
Speech Clarity	An estimated score from 0 to 100 and the improvement, with notes on what was detected. It is VoiceCleaner's own metric, not a standardized intelligibility score.

LIVE PREVIEW OF CHANGES

After the first Enhance, changing a setting re-processes about 10 seconds around the playhead (or your selection, up to 30 s) a moment after you stop adjusting. The previewed part is marked `LIVE PREVIEW` on the Enhanced track and plays at the level a full Enhance would give it. `Enhance Again` applies the settings to the whole recording; exporting first asks whether to do that.

Live preview is on by default with an NVIDIA GPU (about a second per preview) and on Apple Silicon Macs (under half a second, or about 5 s with Voice Re-synthesis). On other computers it can be turned on with the `Live preview of changes` checkbox, but it is slower.

EDIT TAKE

Click `✂ Edit Take…` above the Original track (or right-click a track, or press `E`) to cut out a pop, a click or a stumble. Select it, then **Cut** it (`X` or `Delete`, crossfaded) or **Silence** it (`S`, keeps the timing). Edits apply to both tracks, so A/B still compares the same audio, and they are used when you export. The original file is never changed; the edits are kept with the session and shared with the Compile tab. `Clear Edits` removes them.

7. LUFS (Loudness Units relative to Full Scale) measures perceived loudness over the whole recording, per ITU-R BS.1770-4. -16 LUFS is the usual level for podcasts and online video; broadcast uses -23 (Europe) or -24 (US).
8. dBTP is decibels relative to full scale, measured as "true peak" (with oversampling), which catches peaks a plain sample meter misses. A ceiling of -1 dBTP leaves room for MP3 and streaming encoders.

UNDO AND REDO

Settings changes can be undone with `Ctrl`+`Z` and redone with `Ctrl`+`Y` or `Ctrl`+`Shift`+`Z` (on a Mac, `⇧ Shift`+`⌘ Cmd`+`Z`), or the ↶ ↷ buttons next to the preset. A slider drag is one step; the button's tooltip names the change.

CHAPTER 06

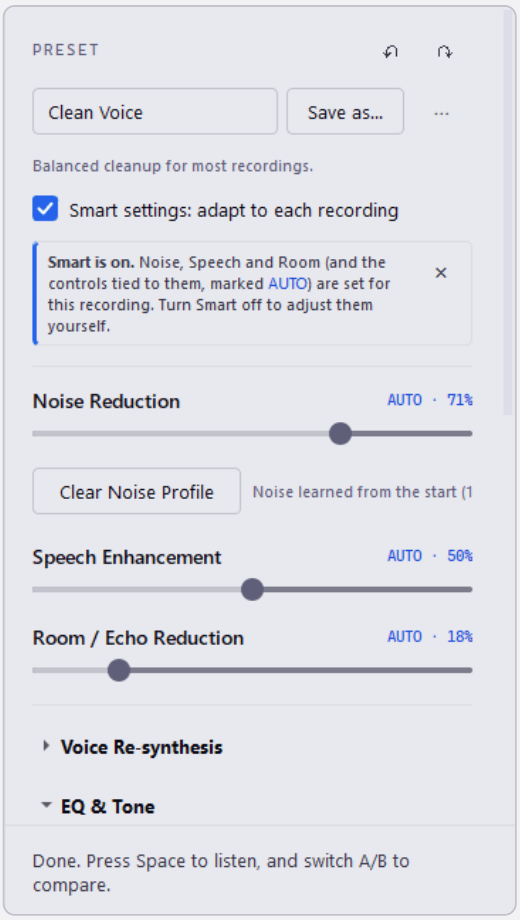
SETTINGS AND PRESETS

A preset and three sliders cover most recordings. Everything else is there when you need it.

The sidebar is the same in the Enhance and Batch tabs. Every control is remembered between sessions. The mouse wheel scrolls the sidebar, even over a slider; click a control first to adjust it with the wheel. Double-click any slider to reset just that one.

PRESETS

A preset is a starting point. `MODIFIED` appears when you change a control; `Reset` goes back to the preset.



The top of the settings sidebar.

PRESET	USE IT FOR
Natural	Minimal processing. Keeps the room and the voice as they are, just cleaner.
Clean Voice DEFAULT	Balanced cleanup for most recordings.
Noisy Recording	Strong noise reduction for fans, traffic and busy rooms.
Interview	Moderate noise reduction, clearer speech, even levels between speakers.
Podcast	Broadcast-style voice: enhanced, gently compressed, consistent.
Extreme Noise	Maximum AI enhancement and voice re-synthesis with artifact protection.

Your own presets: `Save as…` stores all current settings as a preset. The `⋯` next to it updates, renames or deletes it, or shows the presets folder. Presets are small JSON files, easy to back up or share.[9]

SMART SETTINGS

`Smart settings: adapt to each recording` is on by default. The analysis sets Noise, Speech and Room, and the controls tied to them, for each recording: clean recordings get light processing and noisy ones get more. While Smart is on, those controls are locked and marked AUTO, and a note explains why (the × hides the note; `Show Tips Again` brings it back). Turn Smart off to set them yourself; your values then stay as they are when you open another recording. Re-synthesis, leveling, EQ and the rest are always yours.

THE THREE MAIN SLIDERS

Noise Reduction

Removes background noise such as fans, hiss, hum and room noise.

Speech Enhancement

Improves clarity: balances the tone of the voice and brings out presence.

Room / Echo Reduction

Reduces the sound of the room (reverb) and echoes.

VOICE RE-SYNTHESIS

Re-generates the cleaned voice with a neural vocoder (NVIDIA BigVGAN). It smooths away leftover processing artifacts and restores natural harmonics. Off by default; 20–40% polishes the voice, while high amounts can change its character. The model (490 MB) is downloaded once from inside the app with `Download voice model`, then works offline.[10]

EQ & TONE

A live response curve, `Tonal balancing` (how much automatic correction toward a natural voice spectrum), `Tilt` (darker / brighter), and eight band sliders from 80 Hz to 10 kHz, ±12 dB. Double-click a band to reset it. If the EQ cuts the body of the voice (250 Hz–1 kHz) by 9 dB or more, a note warns that it can sound hollow or distant.

DYNAMICS

- **Leveling** (amount, maximum correction, speed) evens out sentences and speakers over seconds, and holds still in pauses.
- **Compression** (amount, threshold, ratio, attack, release) controls word-level peaks.
- **De-esser** (50% by default) softens harsh “s”, “sh” and “t” sounds only at the moments they stick out.
- **Pauses**: `Shorten long pauses`, off by default. See Shortening pauses.

9. Windows: `%APPDATA%\VoiceCleaner\presets`. Mac: `~/Library/Application Support/VoiceCleaner/presets`.
10. Speed: about real time on a CPU (a 10-minute recording takes about 10 minutes more), about 3× real time on an Apple Silicon GPU, and much faster on an NVIDIA GPU.

ADVANCED

GROUP	CONTROLS
Noise reduction	AI speech enhancement (on/off), Repair clipping, amount, noise floor estimate, spectral subtraction strength, noise gate threshold, residual noise floor, pause clean-up
Speech enhancement	Enhancement strength, voice presence (around 3 kHz), high-frequency clarity (above 7 kHz), low-frequency cleanup
Room and echo	Reverb reduction, echo reduction
Loudness	Normalize loudness, target loudness (-30 to -10 LUFS, default -16), peak ceiling (default -1 dBTP)
Detected automatically	What the analysis found: background noise and speech-to-noise, hum, room reverb, echo, voice pitch, high-pass filter and stereo layout; and `Reset to preset`

> **TIP: LEAVE HEADROOM FOR NATURAL SOUND**
>
> `Residual noise floor` sets how far the background may be reduced. Keeping a little room tone sounds more natural than digital silence, which is why VoiceCleaner never removes all of it by default.

THE QUALITY CHECK

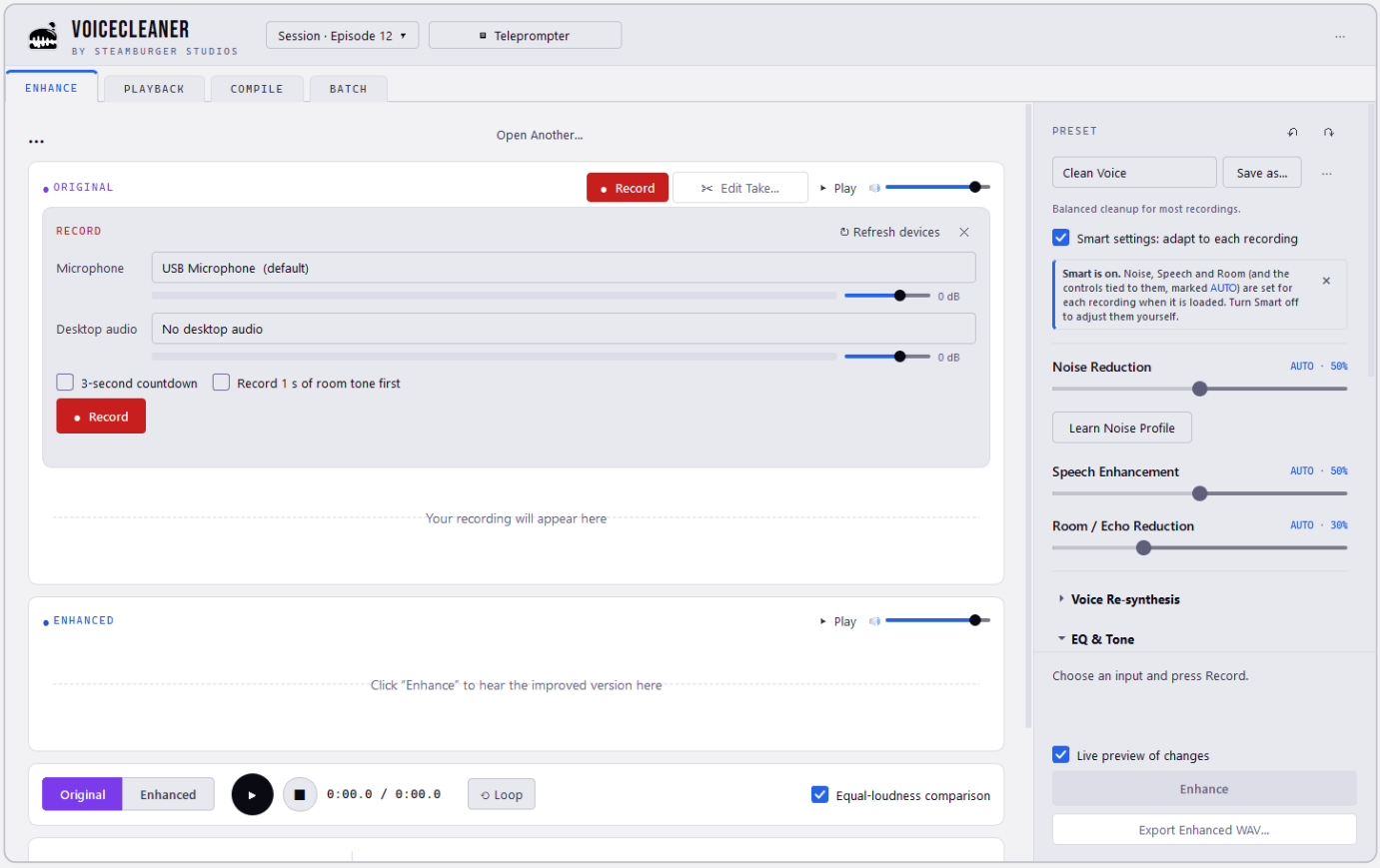
Every result goes through a quality check that compares it with the original: lost syllables, a changed voice tone, high-frequency artifacts, over-compression, clipping, and pauses that came out louder than in the original. If it finds a problem, it reprocesses with gentler settings. The guiding rule is better speech, not obviously processed speech.

CHAPTER 07

RECORDING

Record a microphone, the computer's own sound, or both at once.

Click `● Record Audio` on the start screen, `● Record` on the Original track, or press `R`. The record panel opens above the Original waveform.



▸ The record panel: a microphone and/or desktop audio, with level meters, trims and the recording options.

CHOOSING WHAT TO RECORD

- **Microphone**: any USB microphone, headset or line-in.
- **Desktop audio**: anything playing through a chosen speaker or headphone output, so no “Stereo Mix” is needed.[11]
- **Both at once** captures both sides of a call. The two are aligned and corrected for clock drift.

Each source has a level meter and a trim. `↻ Refresh devices` picks up a microphone you just plugged in.

OPTIONS

- **3-second countdown** before recording starts. Click Record again to cancel it.

11. On Windows this uses WASAPI loopback, which captures what an output device is playing. On a Mac it uses System audio on macOS 14.2 or later, or a loopback driver such as BlackHole on older versions.

- **Record 1 s of room tone first**: recording starts with a quiet second (Stay quiet, then Speak now), and that second becomes the noise profile, so the background is measured exactly.

Both are remembered between sessions.

RECORDING WITH A FILE ALREADY OPEN

MODE	WHAT HAPPENS
New recording	The take replaces the Original.
Append	The take is added to the end of the Original.
Overdub	The take is layered onto the Original from the playhead, optionally while the Original plays (use headphones). Latency is compensated.

WHERE TAKES GO

Takes are saved to the session's `Recordings` folder as 48 kHz 24-bit WAV, and streamed to disk while recording, so an unplugged microphone or a crash keeps what was captured. Combined results (append, overdub) are written as new files; the original is never changed.

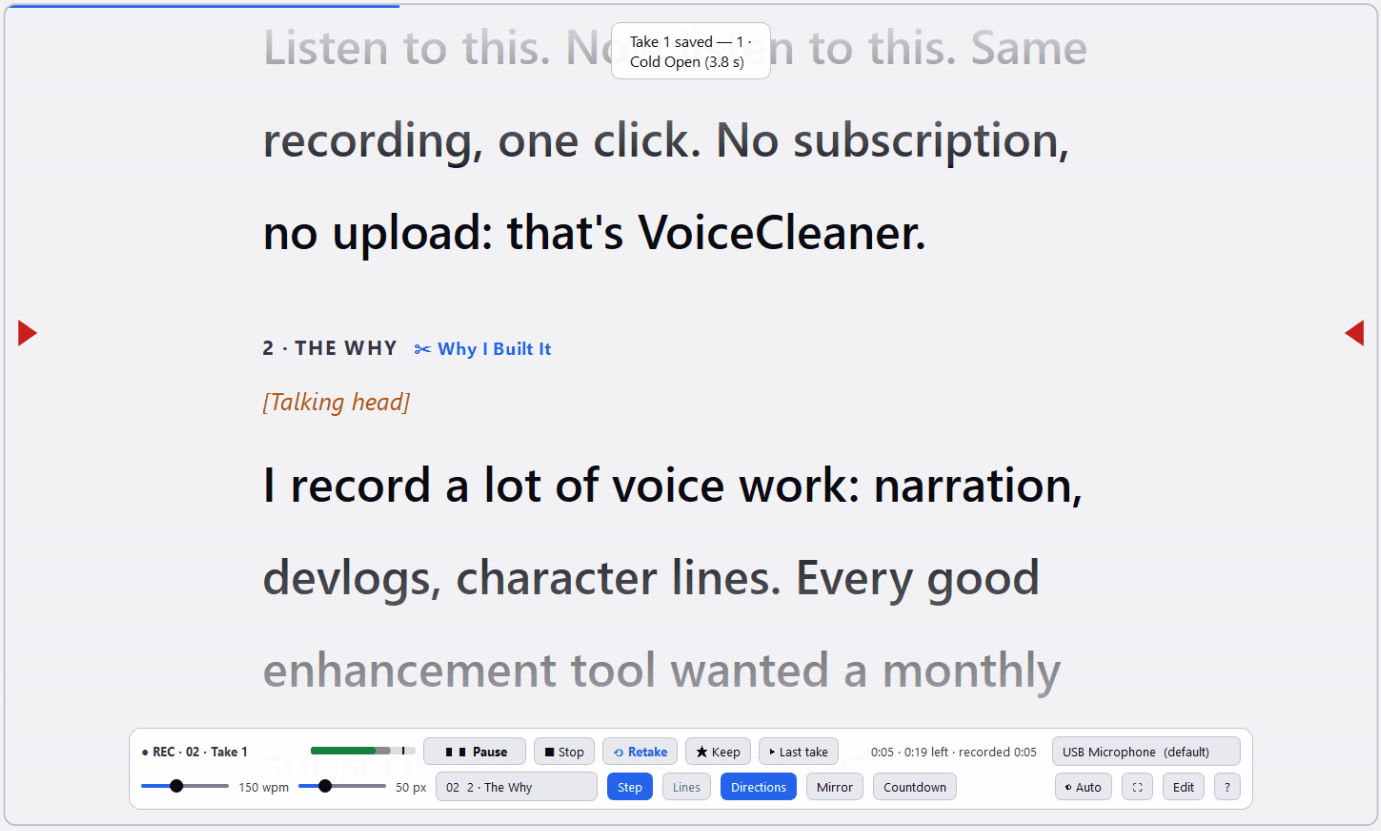
> **NO SOUND FROM THE MICROPHONE?**
> Check that the operating system allows apps to use it: `Settings → Privacy → Microphone` on Windows, `System Settings → Privacy & Security → Microphone` on a Mac.

CHAPTER 08

THE TELEPROMPTER

Read your script while VoiceCleaner records, one take per section.

Open it with `Ctrl`+`T`, the `Teleprompter` button in the header, or `Teleprompter` on the start screen. Every take is kept and enhanced in the background with the current settings, and the Compile tab joins them in script order.



Recording from the teleprompter: the reading line, the REC indicator with take number, level meter and time.

WRITING THE SCRIPT

Paste or edit the script in the prompter (`Edit`, `E`), or open a `.txt` or `.md` file.

```
## 1 · Cold Open | The Hook
[B-roll: A/B toggle, noisy waveform snapping clean]
Listen to this. Now listen to this. Same recording, one click.

## 2 · The Why | Why I Built It
I record a lot of voice work: narration, devlogs, character lines.
```

- `## Heading | Clip name` starts a section. The clip name names its takes.
- `[brackets]` are stage directions; `D` shows or hides them.
- A blank line starts a paragraph.

RECORDING

`Start` (`Space`) counts down 3-2-1 and records from the chosen microphone without a break. When the next section reaches the reading line (or you press `→` / `Page Down`), a new take starts by itself. The cut is placed in the pause between the sections, so a word that runs over is never cut or repeated.

- **Retake** (`Backspace` or `Page Up`) goes back to the start of the section and records it again. Every take is kept; the newest is used unless you mark one with **Keep** (`G`).
- **Last take** (`P`) plays back the most recent take.
- **Step** mode holds each section (or line) until you move on. **Lines** records one take per paragraph instead of per section, for line-by-line voice work.
- Presentation clickers and foot pedals that send Page Up / Page Down work hands-free.
- If the microphone sends only silence, a `NO SOUND` warning names it.

TELEPROMPTER KEYS

`Space`	Start / pause (pausing pauses the take)	`↑` `↓`	Speed (words per minute)
`→` `PgDn`	Next section	`+` `−`	Text size
`←`	Previous section	`M`	Mirror (for teleprompter glass)
`⌫` `PgUp`	Retake	`D`	Stage directions on/off
`G`	Keep this take	`F`	Fullscreen
`P`	Play last take	`E`	Edit the script
`Home`	Back to the top	`H`	Show the key help
`Esc`	Stop recording / leave fullscreen		

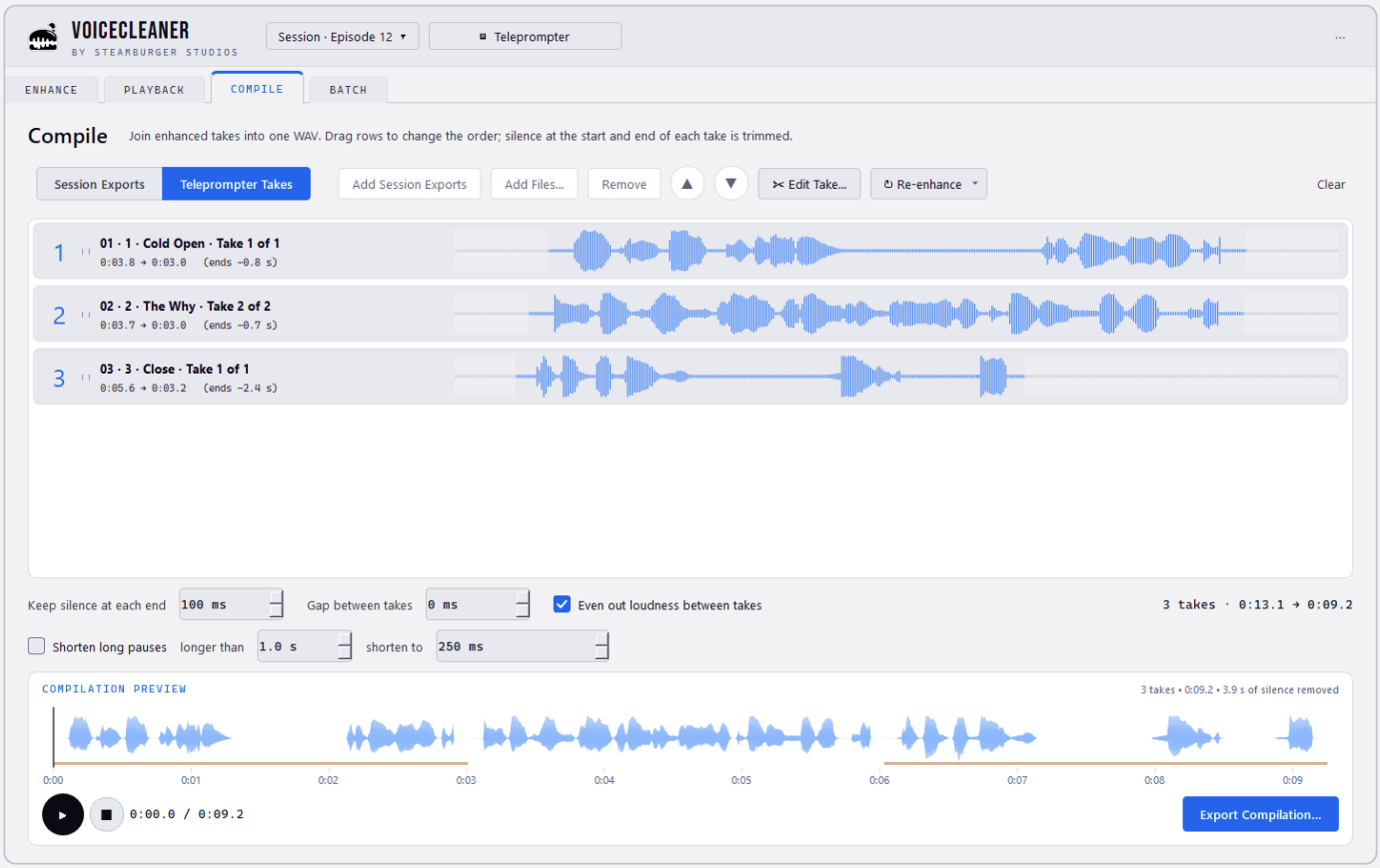
The prompter's colours follow the app's day/night setting; the ◐ button keeps it always Dark or Light instead.

Takes are saved in the session's `Recordings` folder, named after the section, for example `Episode 12 - 03 What It Does - Take 2.wav`, enhanced into `Exports`, and listed in Compile.

CHAPTER 09

THE COMPILE TAB

Join enhanced takes into one finished WAV.



Compile with the teleprompter's takes in script order and the compilation preview.

BUILDING THE LIST

- The list starts with the session's enhanced exports (`Session Exports`). Add more with `Add Files…` , by dropping files on the window, or with `Add to Compilation` in the Playback tab.
- `Teleprompter Takes` shows the script instead: one row per section, in order, with its selected take (enhanced when ready). Right-click a row to use another take; it then counts as that section's keeper.
- Drag rows to reorder, or use `▲` / `▼` or `Ctrl`+`↑` / `Ctrl`+`↓`. Double-click a row to hear that take alone.

OPTIONS

- **Keep silence at each end**: leading and trailing silence of every take is trimmed, keeping up to 100 ms (adjustable) so breaths and soft onsets survive. Short fades prevent clicks.[12]
- **Gap between takes**: optional silence between takes.

12. 5 ms fades at each cut. Short enough to be inaudible, long enough to avoid the click a hard cut makes.

- **Even out loudness between takes**: matches the takes' loudness. A true-peak limiter keeps the result from clipping.
- **Shorten long pauses**: see below.

The preview updates itself a moment after each change, so ▶ starts at once. Reordering while it plays keeps playing at the same take.

EDITING AND RE-ENHANCING TAKES

- ✂ **Edit Take…** (`E` or right-click): select a pop, click or a pause the trimming missed and **Cut** it (`X` / `Delete`) or **Silence** it (`S`). `Ctrl`+wheel zooms down to a few milliseconds; `Hear Result` plays the take as it will sound. Edits are saved in the session (`compile_edits.json`); the file itself is never changed.
- ↻ **Re-enhance** (select several rows to do them all) enhances takes again from their original recordings, with the current settings or a preset, in the background. The replaced version is kept in `Exports\Previous Versions`; `Revert to Previous Version` swaps it back. Take edits stay in place.
- **Open in Enhance Tab** loads one take's recording for fine-tuning; `Update Take in Compile` puts the result back.

EXPORTING

`Export Compilation…` saves to the session's `Exports` folder as `<Session> - Compilation.wav`.

SHORTENING PAUSES

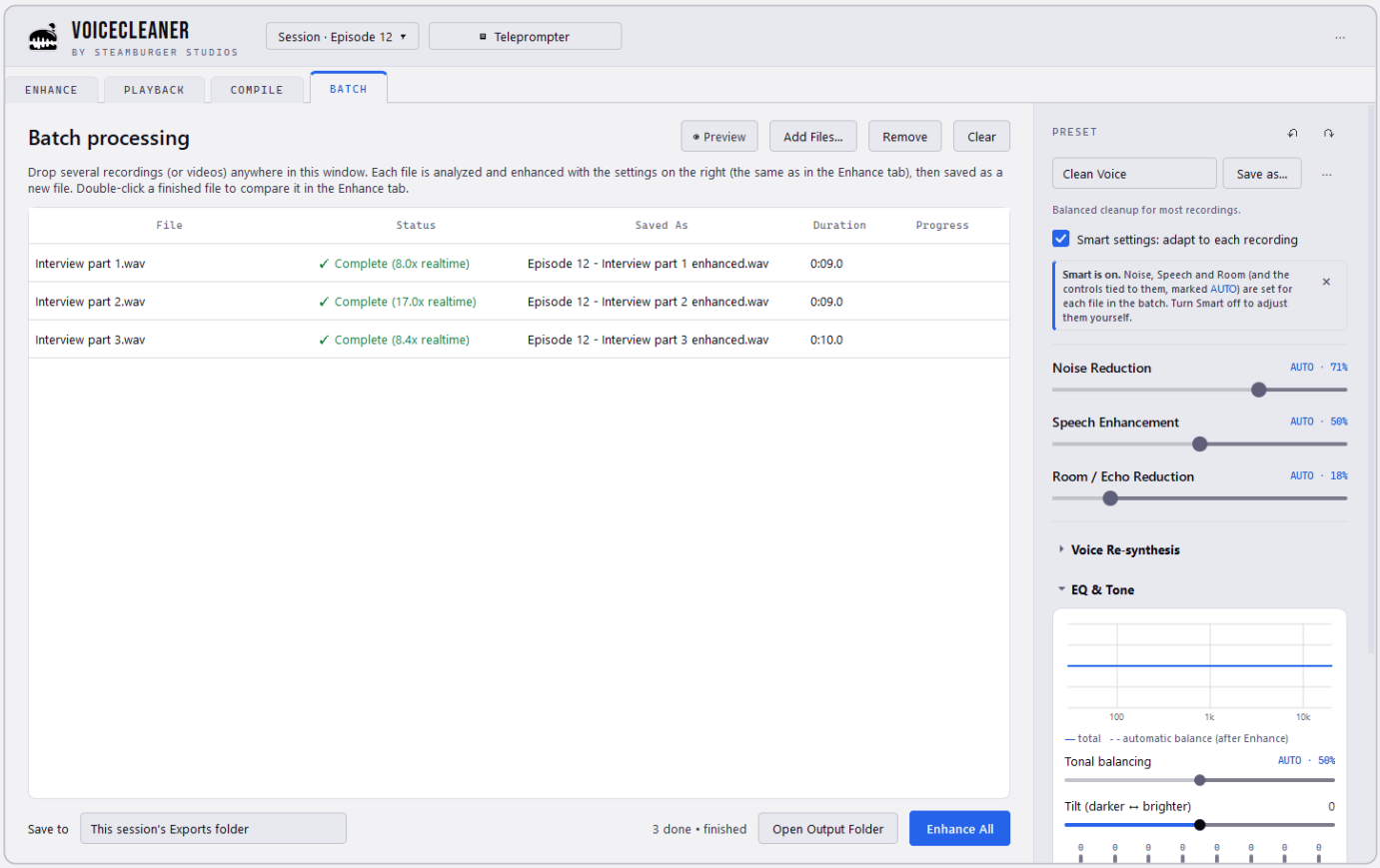
Optional and off by default. Pauses longer than a threshold (1.0 s by default, 0.3–5 s) are shortened to a natural pause (250 ms by default, up to 2 s). The kept pause is made of the start and end of the original silence, joined by a short crossfade, so breaths survive and the background continues without clicks. Setting the length to 0 ("Remove completely") removes long pauses entirely. Silence before the first word and after the last is left to trimming. It works in two places:

- **Compile tab**: applied to every take before joining; rows mark the shortened pauses and the preview includes them.
- **Enhance tab** (Dynamics → Pauses): applied when exporting, including Batch. The preview keeps the original timing, so A/B stays aligned. A note says how many pauses will be shortened and the time saved. Presets never turn it on.

CHAPTER 10

THE BATCH TAB

Enhance a whole folder of recordings with the same settings.



Batch processing of three files, with the settings sidebar on the right.

1. Drop several files (or a folder) anywhere in the window, or click `Add Files…`. Recordings in any supported format and videos work. `Remove` and `Clear` tidy the list.
2. Set up the sidebar on the right: it is the same as in the Enhance tab, and Smart settings still adapts to each file.
3. Optional: turn on `◉ Preview` (`P`) and click a file to hear a 10-second sample of it (the part with the most speech) with the current settings. Compare it A/B at equal loudness (`Space`, `A`, `B`). The sample follows setting changes, so you can tune before enhancing everything.
4. Choose `Save to` : This session's Exports folder, Same folder as each original, or Choose folder…
5. Click `Enhance All`.

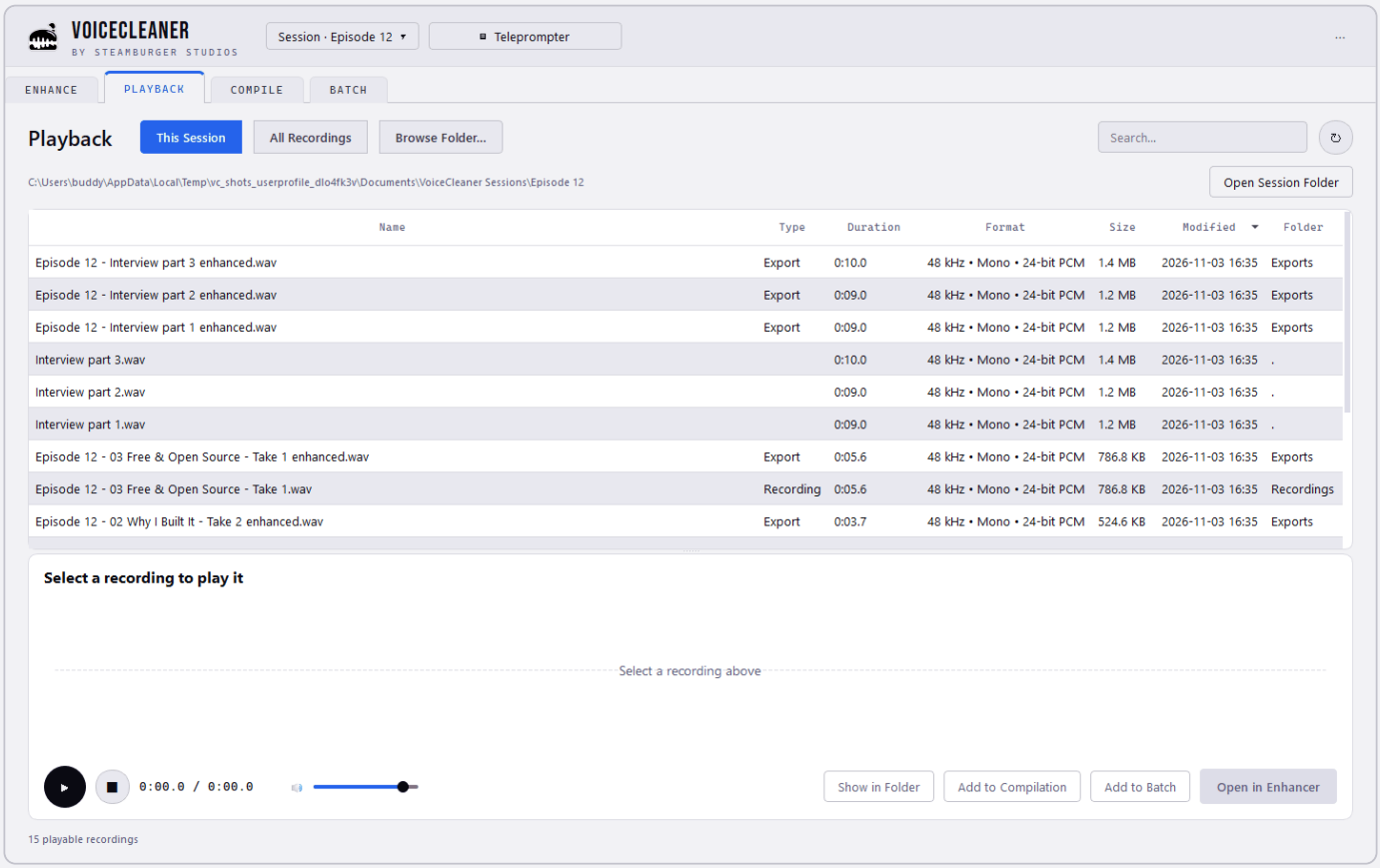
Each file is analyzed and enhanced on its own. A damaged file is reported and the rest continue. When it is done, `Open Output Folder` shows the results, and double-clicking a finished file opens it in the Enhance tab to compare. Results never overwrite an original. The `Saved As` column names each result.

To run batches without the window, for example from a script, see the command line.

CHAPTER 11

THE PLAYBACK TAB

Find, play and route your recordings.



The Playback tab listing the current session's recordings.

- `This Session` lists the current session's recordings and exports.
- `All Recordings` lists every session in the default recordings folder.
- `Browse Folder…` scans any folder, including subfolders. Recordings and videos that can actually be played are listed; damaged ones are counted and skipped.

Search, sort, and play with a waveform and seek. Then send the selection on with `Open in Enhancer`, `Add to Batch`, `Add to Compilation` or `Show in Folder`.

CHAPTER 12

EXPORT FORMATS

WAV for editing and archiving, MP3 for sharing.

OPTION	CHOICES
Format	**WAV** (recommended) or MP3
Bit depth (WAV)	16-bit (dithered[13]), **24-bit** (default), or 32-bit float
MP3 quality	High (VBR ·190 kbps), Standard (·130 kbps), Small file (·100 kbps)
Sample rate	Same as original, or 44.1, 48 or 96 kHz

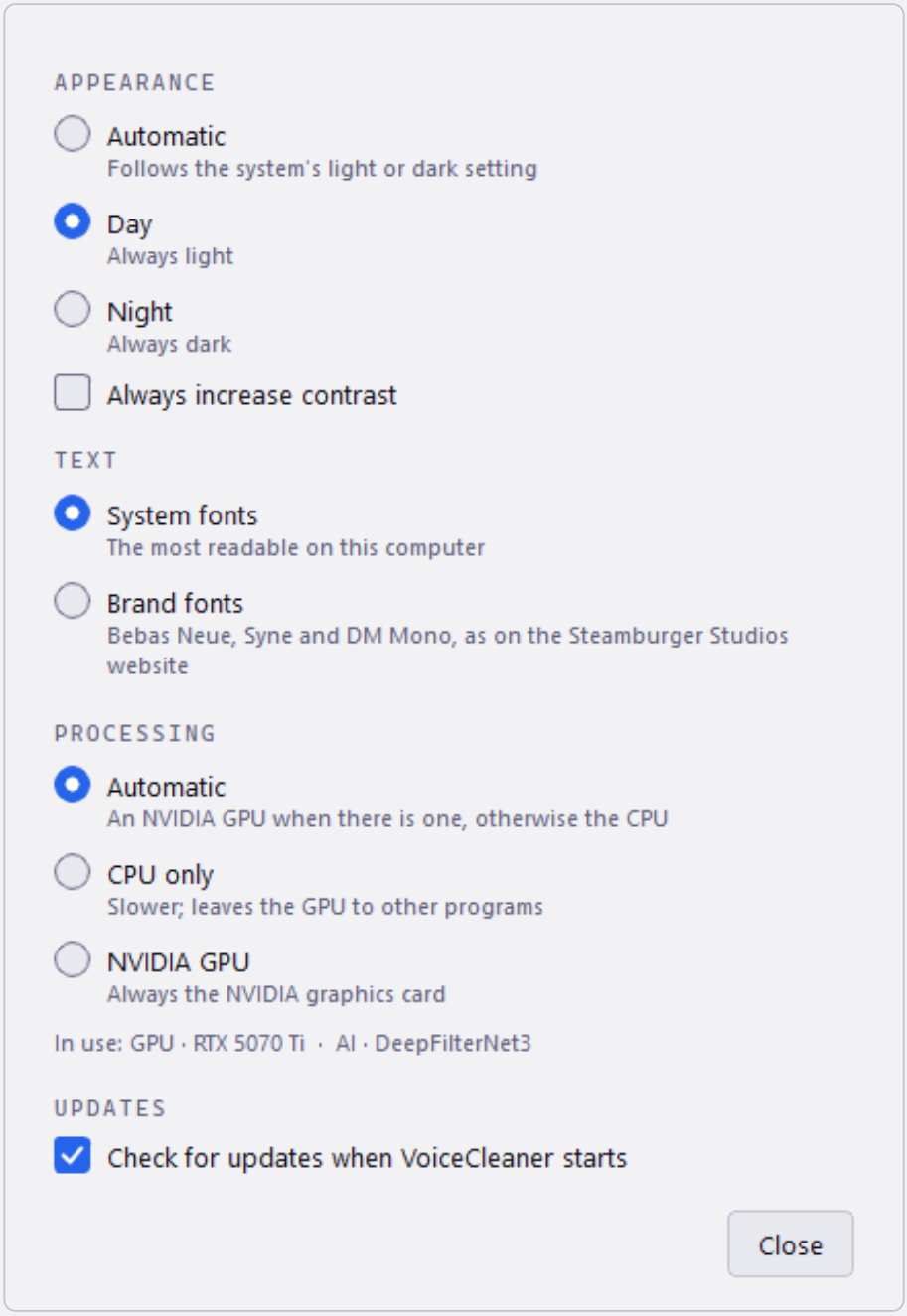
Exports are named after the original (also for phone memos and videos), which is never overwritten. Tick `Show the file in File Explorer when done` (Finder on a Mac) to go straight to the file; the choice is remembered.

13. Dither adds a tiny, inaudible noise when reducing to 16 bits, which avoids the gritty distortion of plain truncation on quiet passages and fades.

CHAPTER 13

SETTINGS

How VoiceCleaner looks and runs. Open it with `⋯ → Settings…` or `Ctrl`+`,` (on a Mac, `VoiceCleaner → Settings…`, `⌘ Cmd`+`,`). Changes apply at once.



The Settings window.

DAY AND NIGHT

VoiceCleaner follows the system's light/dark setting (`Automatic`). Choose `Day` or `Night` to keep one. Colours follow the Steamburger Studios website.

CONTRAST

When the system asks for more contrast (a Windows contrast theme, or Increase Contrast on a Mac), secondary text, borders and slider tracks become stronger. `Always increase contrast` does the same whatever the system setting.

FONTS

System fonts are used by default for readability. `Brand fonts` switches to the Steamburger Studios type: Bebas Neue, Syne and DM Mono.

PROCESSING DEVICE

`Automatic` uses an NVIDIA GPU when there is one (on a Mac, the Apple GPU for re-synthesis); `CPU only` forces the processor. In use shows what processes the audio now, and the AI model.

UPDATES

`Check for updates when VoiceCleaner starts`; see Updates.

AUDIO OUTPUT

`⋯ → Audio Output` selects `System default` or a specific output for every playback in the app, including take previews. Active outputs are listed first; disconnected or disabled ones are under `Inactive`. The choice is remembered. If a selected device is disconnected, playback pauses and the selection is kept: reconnect it and press Play, or choose another output. Finish recording before changing outputs or refreshing devices.

CHAPTER 14

COMMAND LINE

The Batch tab's processing, without the window.

On Windows, `VoiceCleanerCLI.exe` is next to `VoiceCleaner.exe` (by default in `C:\Program Files\Steamburger Studios\VoiceCleaner`).

```
VoiceCleanerCLI --batch recordings out --preset Natural
VoiceCleanerCLI --batch "Voice Memo 14.m4a" zoom_0.mp4 out --format mp3
VoiceCleanerCLI --batch D:\Podcast --recursive D:\Podcast\clean --preset Podcast --no-smart
VoiceCleanerCLI --list-presets
```

OPTION	MEANING
`--batch IN… OUT`	Inputs are files and/or folders; the last path is the output folder. Results are named `<name>_enhanced.wav` and never overwrite anything.
`--recursive`	Include subfolders.
`--preset NAME`	A built-in preset or one of your own.
`--no-smart`	Use the preset as it is, without adapting it to each file.
`--format wav|mp3`	Output format.
`--bits 16|24|32`	WAV bit depth (32 = float).
`--device auto|cpu|cuda`	Processor to use.
`--list-presets`	List the available presets.
`--help`	List everything.

It prints one line per file with progress and a summary at the end. The exit status is `0` when all files were enhanced, `1` when some failed (the others are still done), and `2` when there was nothing to do or the command had a mistake.

CHAPTER 15

KEYBOARD SHORTCUTS

Press `Ctrl`+`/` in the app for this list. On a Mac, `Ctrl` is `⌘ Cmd`.

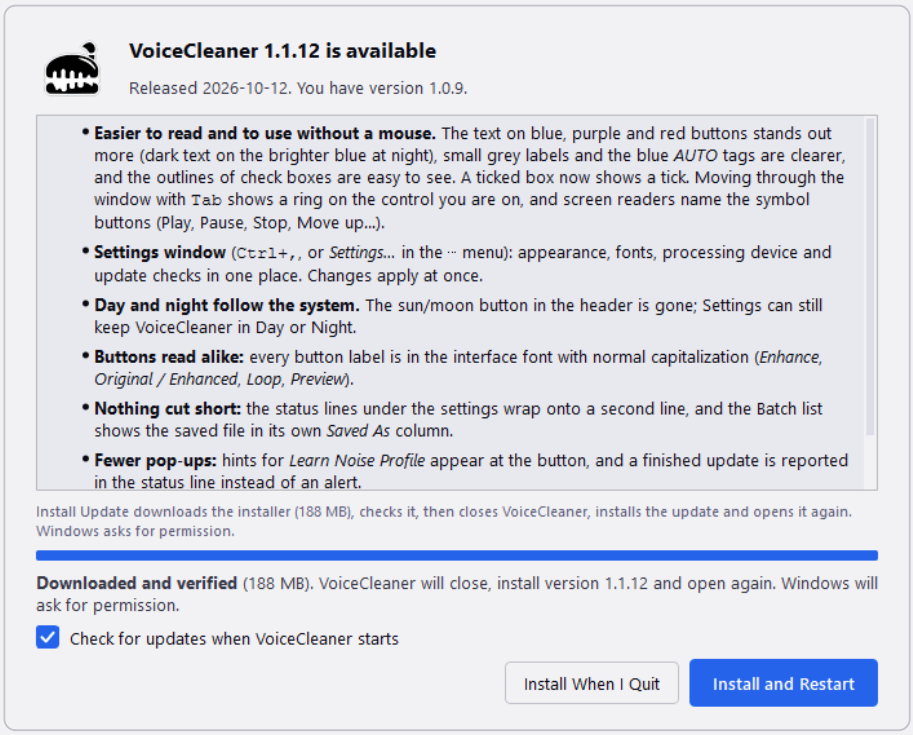
`Space`	Play / pause (the tab in front)	`Ctrl`+`O`	Open a recording or video
`A` / `B`	Original / enhanced	`Ctrl`+`Enter`	Enhance
`Home`	Back to the start	`Ctrl`+`E`	Export enhanced audio
`L`	Loop selection / recording	`Ctrl`+`Shift`+`N`	New session
`R`	Record / stop	`Ctrl`+`T`	Teleprompter
`E`	Edit take	`Ctrl`+`Z`	Undo a settings change
`X` / `Del`	Cut (in the take editor)	`Ctrl`+`Y` (Mac: `⇧` `⌘` `Z`)	Redo
`S`	Silence (in the take editor)	`Ctrl`+`/`	Shortcut list
`P`	Batch preview on/off	`Ctrl`+wheel	Zoom the waveform
`Esc`	Cancel	Double-click	Reset a slider
`Tab`	Next control (a ring shows where you are)	`Ctrl`+`,`	Settings
		`F1` (Mac: `⌘` `?`)	This user manual

The teleprompter has its own keys; see Chapter 8, or press `H` in it.

CHAPTER 16

UPDATES

VoiceCleaner checks for new versions, and installs them for you on Windows.



An update downloaded and verified, ready to install.

`··· → Check for Updates…` asks GitHub for the latest release. If there is a newer version, the app shows what is new. Nothing is downloaded or installed until you choose to, and nothing about you, your computer or your audio is sent.

- **Installed on Windows:** `Install Update` downloads the installer inside the app (an interrupted download continues where it stopped), checks it against the checksum GitHub publishes,[14] then offers `Install and Restart` or `Install When I Quit`. Windows asks for permission, the installer runs, and VoiceCleaner reopens and confirms the update.
- **Portable copies:** `Download` opens the release page.
- **Mac:** `Download` fetches and checks the DMG, then shows it in Finder; drag the new VoiceCleaner to Applications.

By default the app also checks once each time it starts and only speaks up when there is something new. `Later` hides that version until the next day; `Skip Version` hides it for good. Turn automatic checks off with the checkbox in the update window.[15] After an update, a short `What's New` tour shows only what changed.

14. A SHA-256 checksum is a fingerprint of the file. If even one byte differed from the published release, the check would fail and the update would not be installed.

15. Administrators can disable automatic checks entirely by setting the environment variable `VOICECLEANER_NO_UPDATE_CHECK=1`. To test the connection from a terminal: `VoiceCleaner --check-updates`.

CHAPTER 17

REFERENCE

Supported input

- **Opened directly:** WAV (including WAVE_FORMAT_EXTENSIBLE, RF64 and W64), MP3, FLAC, Opus and Vorbis (OGG), AIFF and CAF.
- **Through the included FFmpeg:** M4A and AAC (phone voice memos), WMA, AMR, and the sound of video files (MP4, MOV, M4V, MKV, WebM, AVI, WMV, 3GP). The first audio track is used; surround is mixed down to stereo.
- Mono or stereo, 8 kHz to 384 kHz, up to 6 hours per file. Recordings too large for memory are processed from disk automatically.

STEREO CONTENT	HOW IT IS PROCESSED
Identical channels	Processed once as mono, written back as stereo
Speech on one side only	That channel is processed, then centred on both
Genuine stereo	Each channel processed, with linked dynamics

How VoiceCleaner cleans speech

PROBLEM	HOW IT IS HANDLED
Steady noise (fans, HVAC, hiss)	A noise profile learned from pauses (or your selection), a Wiener filter, then the AI model
Changing noise (traffic, rustle, babble)	DeepFilterNet3 neural speech enhancement
Clipping (recorded too loud)	The cut-off peaks are rebuilt; every unclipped sample is kept
Clicks, taps, keyboard in pauses	A transient suppressor that never touches speech
Electrical hum	Automatic 50/60 Hz and harmonic detection, narrow notches
Rumble	Adaptive high-pass, always below the speaker's lowest pitch
Room reverb, mild echo	Late-reverb suppression using the measured reverb time;[16] echo detection
Dull, muddy or boomy voice	Adaptive EQ toward a natural long-term speech spectrum
Leftover artifacts, thin voice	Optional neural voice re-synthesis (NVIDIA BigVGAN-v2)
Uneven levels	Speech-aware leveling, a gentle compressor and an optional soft expander

16. The reverb time (RT60) is how long the room takes for a sound to die away by 60 dB. VoiceCleaner measures it from the recording after removing noise.

PROBLEM	HOW IT IS HANDLED
Loudness	-16 LUFS by default, true-peak limiter at -1 dBTP

Files and folders

WHAT	WINDOWS	MAC
Sessions	`Documents\VoiceCleaner Sessions\<session>\`	`~/Documents/VoiceCleaner Sessions/<session>/`
Preferences	`%APPDATA%\VoiceCleaner\config.json`	`~/Library/Application Support/VoiceCleaner`
Presets	`%APPDATA%\VoiceCleaner\presets`	`~/Library/Application Support/VoiceCleaner/presets`
Logs	`%LOCALAPPDATA%\VoiceCleaner\logs`	`~/Library/Logs/VoiceCleaner`
Re-synthesis model	`%LOCALAPPDATA%\VoiceCleaner\models`	`~/Library/Application Support/VoiceCleaner`
Downloaded updates	`%LOCALAPPDATA%\VoiceCleaner\updates`	Shown in Finder after download

The log records file names, settings, devices, timings and errors, never audio content. `⋯ → Open Log Folder` opens it.

CHAPTER 18

TROUBLESHOOTING

Windows says “Windows protected your PC”.

The installer is not code-signed yet. Choose `More info` → `Run anyway`.

macOS won't open VoiceCleaner the first time.

Confirm opening it when macOS asks. If it only offers to move it to the Bin, open `System Settings → Privacy & Security` and choose `Open Anyway`.

The recorder shows NO SOUND or flat meters.

Allow microphone access in the system privacy settings (see Recording), check the trim, and click `↻ Refresh devices` after plugging in a microphone.

Playback stopped and an output is shown as unavailable.

The selected output was disconnected. Reconnect it and press Play, or pick another one in `⋯ → Audio Output`.

An M4A or video file won't open.

VoiceCleaner opens them with the FFmpeg it includes. If one still fails, the file may be damaged or use the sound of a DRM-protected video; the log (`⋯ → Open Log Folder`) names the reason. Converting it to WAV in another app works around it.

Enhancing is slow.

On a CPU, processing (and especially Voice Re-synthesis) takes longer. Use the GPU edition on a PC with an NVIDIA card, check `Settings → Processing device`, and keep Re-synthesis at 0% unless you need it.

The header says the AI model is not installed.

VoiceCleaner still works with its own spectral noise reduction. Reinstalling restores the model; `⋯ → Models and Licenses` shows what is installed.

Export asks “Your settings changed since the last Enhance”.

You changed a setting after enhancing. Choose to enhance again so the export matches what you hear in the preview.

The voice sounds thin, hollow or “underwater”.

Lower Noise Reduction or try the **Natural** preset, raise `Residual noise floor`'s room tone, and check the EQ warning for cuts to the 250 Hz–1 kHz range.

Something else went wrong.

Open `⋯ → Open Log Folder` and attach `voicecleaner.log` to an issue on GitHub. The log contains no audio.

KNOWN LIMITATIONS

- De-reverberation is statistical: it clearly reduces room tails but cannot make a very reverberant room sound fully dry.
- Clipping repair rebuilds peaks clipped by a few dB; very hard clipping comes out much better but not like an undistorted recording.
- Speech buried under very loud noise cannot be fully recovered. The quality check favours keeping speech over removing the last bit of noise.

CHAPTER 19

LICENSE AND CREDITS

VoiceCleaner is free and open source under the MIT License. Source code, releases and issues: github.com/BuddyDoQ/VoiceCleaner.

The Steamburger Studios name and burger logo are trademarks of Steamburger Studios and are not covered by the MIT License. Bundled third-party components keep their own licenses: DeepFilterNet3 (MIT or Apache-2.0), NVIDIA BigVGAN-v2 (MIT), PyTorch, Qt for Python (LGPL), and FFmpeg (LGPL build; its source is attached to each release). `⋯ → Models and Licenses` lists them in the app.

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